

Exponential Map

1.1 One-Parameter Subgroups and the Exponential Map

Definition 1.1.1

A *one-parameter subgroup of G* is defined to be a Lie group homomorphism $\gamma : \mathbb{R} \rightarrow G$, with $\mathbb{R}$ considered as a Lie group under addition.

Theorem 1.1.2 ► Characterization of One-Parameter Subgroups

Let G be a Lie group. The one-parameter subgroups of G are precisely the maximal integral curves of left-invariant vector fields starting at the identity.